Introduction to Artificial Intelligence

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Artificial Intelligence (A35728)

Short Title:	Intro to AI		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Mathematics and Physics		
Author	ADAVY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This Module is an introductory Module, which focuses on inspiring students' interest in artificial intelligence. It lays the foundation for students' follow-up study of machine learning, neural network and deep learning, computer vision and pattern recognition, game theory, and application. Through the study of this Module, students will be introduced to the fundamental theories and techniques used in the field of Artificial Intelligence, including searching, knowledge representation, machine learning, and so on. By the end of this course, students will have clear ideas of AI, and some of the major developments of AI of today, and be able to make use of available development tools to explore some areas of application such as iris flowers classification, handwritten digits recognition, and so on.

Programmes

`COMP-0677`	BSc (Hons) in Applied Computing (International) (`WD_KACMI_B`)	1 / 2 / M

Indicative Content

- Brief introduction to birth and history of Artificial intelligence, as well as essential concepts and methods:
- 1. Search methods (deep first Search, broad first search, uniform cost search, greedy search, A* search, graph search)
- 2. Supervised learning (Linear/logistic regression and classification)
- 3. Unsupervised learning (clustering and dimensionality reduction)
- 4. Reinforcement learning (Q-learning)
- 5. Introduction to genetic algorithms and deep learning
- 6. Data preprocessing

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Understand what constitutes ``Artificial'' Intelligence and how to identify systems with Artificial Intelligence.
2. Understand the limitations of current Artificial Intelligence techniques.
3. Familiarity with Artificial Intelligence techniques, such as Searching, Neural Networks, and Machine Learning.
4. Recognize how Artificial Intelligence enables capabilities that are beyond conventional technology, for example, prediction and classification, self-driving cars, and robotics.
5. Ability to apply Artificial Intelligence methods and engineering techniques for solving narrowly defined problems.

Learning and Teaching Methods

- This module will be teaching through a variety of ways, including direct instructions, discussions, and case-based practical studies.
- The topics of Artificial Intelligence and their related concepts will be introduced and explained by instructors in the direct instructions.
- Students in this module are required to engage in discussions and class exercises, thereby enhancing their participation.
- The case-based practical studies implemented in this module will assist students in gaining problem-based skills that result in significant improvement in their critical thinking and active participation in the learning process
- Self-directed learning will also be encouraged throughout the duration of the module.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	30%	1,2
Assignment	70%	1,3,4,5

Assessment Criteria

<40%: Inability to express any knowledge/concepts of Artificial Intelligence. Inability to attend to any discussion. Inability to produce an algorithm to solve a simple problem.

40-49%: Understand basic concepts of certain topics of Artificial Intelligence and able to explain them in their own words. Can recognize search, regression, classification and clustering type of problems.

50-59%: Be able to identify potentially applicable Artificial Intelligence techniques for solving problems. Can develop modeling and make theoretical analysis.

60-69%: All the above and in addition, can explain the differences between different learning methods and make use of their performance measurement metrics.

70-100%: All previous to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	82	

Essential Material

- "Khan Academy." https://www.khanacademy.org/
- "W3 Schools on-line web tutorials." https://www.w3schools.com/
- Russel, and Nerving. _Artificial Intelligence: A Modern Approach_. 4th ed.. : Pearson Education, 2020.

Supplementary Material

- "Python Tutorials." http://introtopython.org
- "Tensorflow." https://www.tensorflow.org/

Requested Resources

- COMPUTER LAB: BYOD Lab